



An underwater habitat in the U.S. Virgin Islands was first placed in 1969.

BIOLOGY

Ear in the Ground

Plants can “hear” rain coming, and it spurs them into action

RAIN SOUNDS MAY BE SOOTHING for humans, but for plants the pitter-patter of droplets is more like a jarring morning alarm. As water falls onto soil or more water, the vibrations are far stronger underground than on the surface, and new research suggests plants take advantage of this wake-up call.

The sound of rain causes rice seeds to sprout faster than they would otherwise, according to a study published in *Scientific Reports*. The results mark the first direct evidence that plants sense and respond to sounds in their environment, the researchers say. And it’s likely that seeds from other plant species behave in the same way.

“It is a little bit of an unusual, quirky effect” that only an interdisciplinary effort between acoustic and plant scientists could have uncovered, says study lead author Nicholas C. Makris, a mechanical engineer who researches acoustics and sensing at the Massachusetts Institute of Technology.

Just how much plants can sense in the world around them has long puzzled researchers. There’s evidence to suggest that plants might have the ability to “[think](#),” “[see](#),” “[hear](#)” and “[make sounds](#),” but scientists have only rarely observed a cause-and-effect relation that reveals our flora friends responding to their environment in real time. In the new study, the researchers home in on the link between sound and germination.

In plants, cellular structures called statoliths are responsible for external sensing. These structures are envelopes of starch that shift and settle at the bottom of plant cells, helping the organisms to detect changes in their position and stability—and to tell germinating seeds in which direction to grow their roots. Makris and study co-author Cadine Navarro, then

OCEANOGRAPHY

Aquanaut Awe

An “underview effect” takes hold on extended seafloor stays

MARK PATTERSON WAS LIVING underwater for a week inside the Hydrolab, a white, cylindrical research station at the bottom of the ocean in the eastern Caribbean. It was 1984, and he was on his first of what would become many missions involving saturation diving: descending to the seafloor and spending multiple days there, leaving the lab during the day to explore the underwater world as an aquanaut. After acclimating to the depths, he couldn’t ascend even if he wanted to. To avoid dire health consequences, he would have to spend 24 hours for every 100 feet of depth slowly decompressing when the mission ended.

Patterson wanted to dive at night. He put on his gear, opened the hatches and swam out into the sea, a 300-foot-

long cord tethering him to the lab. When the cord pulled taut, he sat down on the sandy ocean floor. The lab glowed like a jewel in the distance, and around him bioluminescent plankton shone like stars. “That’s when I felt, ‘Wow, this is the coolest thing maybe I’m ever going to do: live underwater,’” says Patterson, a marine biologist at Northeastern University who has spent a total of 89 days under the sea.

Patterson experienced what scientists have called the “underview effect,” an intense sensation of awe that strengthens aquanauts’ perception of human connectedness to the world. The experience’s name references [the so-called overview effect](#) astronauts describe feeling when looking at Earth from orbit. Patterson was one of the 14 aquanauts who discussed their experiences of awe for a study in *Environment and Behavior*.

Many of those surveyed reported that the length of observation—aquanuts’ daily excursions can last eight hours—is part of what makes the experience special. A moray eel or a barracuda becomes an individual with daily habits and behaviors. Storms overhead

alter pressure below the surface and make ears pop. Plankton undulate with the movement of the waves.

Previous studies have concluded that the overview effect helped astronauts and the people who heard their stories become more attuned to how human behavior has altered Earth. The new study’s authors suggest that descriptions of awe-inspiring time underwater could help others think differently about the seas.

Inducing awe is “one of the strongest ways to weaken the boundaries of ourselves,” says Stanford University psychologist Johannes Eichstaedt, who has studied the overview effect and was not involved in the recent paper. It can generate a sense of connection to nature for some people, he adds.

Lead study author Kristen Kilgallen, a psychology Ph.D. candidate at Northeastern, suggests that awe in the natural world can come simply from trying something new that disrupts everyday routines. “You can find exploration rewarding in and of itself, regardless of what you find,” she says. “That’s what keeps you engaged with the world.”

—Susan Cosier

a graduate student in urban studies at M.I.T., theorized that rain sounds underwater might produce large enough vibrations to jostle the statoliths and possibly prompt germination in rice seeds.

The scientists exposed about 8,000 rice seeds submerged in water, their preferred growing condition, to rain sounds. They found that the noise made these seeds germinate between 30 and 40 percent faster than seeds that were kept in quiet but otherwise identical conditions.

“I think this paper is a really significant advance in our understanding of how plants respond to the acoustic energy in their environment,” says Heidi Appel, a biologist at the University of Houston, who was not involved in the new work. “This is the first study of its kind to measure the vibrational impact of raindrops carefully and realistically.”

Researchers, including Appel and the study authors, hope these findings will be an “aha! moment” for future scientists, inspiring them to study plants’ responses to other natural sounds.

—K. R. Callaway



COLLECTIVE BEHAVIOR

Sheepdog Strategy

Herding a small group of “noisy” sheep holds hints for computer algorithms

WATCHING HOURS of “sheepdog YouTube”—competitions where trained dogs shepherd a small number of unpredictable sheep—gave scientists a key insight into how to control chaotic swarms. The trick, they write *in Science Advances*, is to exploit the sheep’s indecisiveness.

Sheepdogs routinely steer and split groups of 50 or more sheep. But in the century-old tradition of sheepdog trials, dogs display their skills by controlling only four or five. Such a small flock is inherently “noisier”—less predictable—and harder to manage than a large herd. Without the safety of numbers, *their individual personalities* can drive them to frequently switch between “Flee the dog!” and “Keep calm and follow the others.” How do sheepdogs successfully steer small groups of erratic sheep?

By analyzing sheepdog trials on YouTube and talking to farmers in Georgia, the researchers discovered the dogs’ two-step strategy: (1) wait for all the sheep to face the right direction as they flip around at random; (2) immediately chase them.

Once the aligned sheep start moving, they will randomly switch between “follow” and “flee” and quickly break their formation. At that point, the dog pauses and waits for them to

orient again. It’s a slow, “delicate dance,” says study co-author Saad Bhamla, a biomolecular engineer at the Georgia Institute of Technology. But through patience and timing, the dog exploits the sheep’s own randomness to maneuver the flock. Bhamla likens it to sailing a ship on a windy day, raising sails only when the wind is heading your direction.

Inspired by the chaotic sheep, the researchers devised an “Indecisive Swarm Algorithm” for robots that keep switching who they follow (either the researchers’ controller or a neighboring bot). This tactic created easier-to-control robots than having the bots either follow only the controller or average the movement of all their neighbors to follow, which eventually dilutes the controller’s signal. In the future, the researchers say, this algorithm could help to organize other noisy collectives, such as a bunch of drones, self-driving cars or even AI agents working together, which could be programmed to swap between following other individuals’ leads and the overarching controller. “Indecisiveness prevents the group from binding up and makes it more pliable,” says Ted Pavlic, a computer scientist at Arizona State University, who was not involved in the study. Noisiness can help avoid deadlocks like those recently seen affecting groups of autonomous cars.

“We tend to think of noise as a problem that makes systems less predictable, less optimized,” says Raphaël Sarfati, a physicist at Goodfire AI, who was also not involved in the work. “[But] we see everywhere that noise, a little bit of noise at least, is really, really good for driving systems toward a better optimum.”

—Rohini Subrahmanyam

Sheepdog trials put trained dogs to work herding small groups of sheep.



Jacquesvanderinter/Getty Images



Humans have adapted to survive extreme heat, cold and everything in between.

EVOLUTION

Culture Power

Humans covered the planet 300 times faster than possible from genetic evolution alone

JUST UNDER 300,000 YEARS from the moment *Homo sapiens* appeared in Africa, the species had encircled Earth, mastering desolate deserts and frozen wastelands and all the temperate climes in between. Throughout this staggering expansion, we seem to have relied surprisingly little on genetic adaptation to fuel our globe-conquering—all eight billion of us together remain less genetically diverse than individual populations of chimpanzees. So how did we do it?

Many scientists point to cultural evolution, the process by which knowledge, customs and technology spread over time. But according to Alex Mesoudi, who studies cultural evolution at the University of Exeter in England, “it’s always been just a vague claim.”

No longer. A recent paper in the *Proceedings of the National Academy of Sciences USA* by Arizona State University evolutionary anthropologist Charles Perreault calculates just how big a boost our capacity for culture might have given the great human takeover. Had we been a typical mammal, forced to adapt primarily through sluggish genetic evolution, Perreault concludes, we would’ve needed 88 million years to attain our current geographic footprint—and we would have split into some 2,200 distinct species in that time.

To arrive at those figures, Perreault compiled range maps for nearly 6,000 mammal species and charted how geographic spread relates to three proxies of evolutionary change for a group with a common ancestor, called a lineage: a lineage’s age, its number of species, and its spectrum of body sizes. Those relationships let Perreault gauge roughly how much “evolutionary work” mammals do to populate a given area.

By this yardstick, humans accomplished in a few hundred thousand years what otherwise would have required tens of millions. Cultural shortcuts meant that, unlike other mammal lineages, we didn’t have to wait an entire generation to adapt via natural selection. “We can just skip that,” Perreault says. A continuous stream of better tools, smarter ideas and more effective practices “really accelerates the pace of evolution.”

Collectively, Perreault found, humans occupy as much terrain as all other mammals combined, far more than any one species. (Gray wolves come closest, covering half as much land.) This makes us seem like the ultimate generalists, but he notes that in another sense we aren’t, really—it’s just that individual groups of people perfected the art of survival in vastly different places. The Kalahari hunter-gatherers of southern Africa might not last long in Greenland, and vice versa for the Inuit.

Mesoudi, who was not involved in the new study, agrees that cultural evolution was probably a key to human success while noting that the secret sauce could instead turn out to be social cooperation, language, individual intellect, or some combination of these factors. Still, he says, the paper is “a nice attempt to quantify something that we often write but don’t actually put any numbers on.”

—Cody Cottier

SCIENCE IN IMAGES

Martian Shadow

A fast-moving wave of darkness is spreading across the Red Planet

VOLCANIC ASH IS CREEPING across the surface of Mars with startling speed.

The European Space Agency's Mars Express mission has released a stunning orbital image (*below*) showing surprising changes in Mars's Utopia Planitia basin, which is thought to be the site of a now vanished sea. Captured by Mars Express's High Resolution Stereo Camera, the image shows

two abutting landscapes of light and darkness, the former made from Mars's modern-day rusted sands and the latter colored by volcanic minerals from the planet's deep past.

A side-by-side comparison with views of the same patch recorded by NASA's Viking orbiters in 1976 reveals a striking spread of that dark coloration. Visible changes to the Martian



ANIMAL COGNITION

Feline Assistance

Cats, unlike dogs and toddlers, help you only when it helps them

POP CULTURE HOLDS that if you're trapped in a well, Lassie will lead the way to a rescue—but if you're stuck with Garfield, you'd better have some lasagna in your pocket. And research suggests such stereotypes aren't far off.

Scientists compared 19 children between 16 and 24 months old with 38 untrained pet dogs and 22 cats, asking a simple question: Who will spontaneously respond when a human appears to need help? In the experiment, a familiar caregiver—the child's parent or the pet's owner—interacted with a sponge before turning away. Then an experimenter hid it in full view of the study subject. Across three trials of decreasing difficulty—when the sponge was unreachable and covered,

then visible but out of reach, then fully reachable—the person searched, repeating, "I can't find it. What should I do?" but never directly addressing the subject.

The study grew out of a broader question about prosocial behavior—why some species help others and some don't, says comparative ethologist and study co-author Melitta Csepregi, who studies animal behavior at ELTE Eötvös Loránd University in Hungary. "To get at that, we compared dogs, cats and toddlers, three species that live closely with humans but differ sharply in their evolutionary histories."

In the findings described in *Animal Behaviour*, all three groups paid similar levels of attention. But children and dogs were more likely to show helping-related behaviors—approaching, indicating or retrieving the object for the person. By the final trial, more than half the dogs and nearly half the toddlers indicated the object's location, and some also brought it to the caregiver. Cats never approached it and only rarely indicated its location.

University of Vienna cognitive biologist Ludwig Huber, who

ESA/DLR/FU Berlin (Mars)

surface are more often marked by millions of years, not by dozens of them, but this isn't the first time observers have witnessed strange waves of darkness spreading on Mars.

According to planetary scientists, the world's strong winds might explain the current spread. Either by blowing around surface deposits of volcanic ash from ancient eruptions or

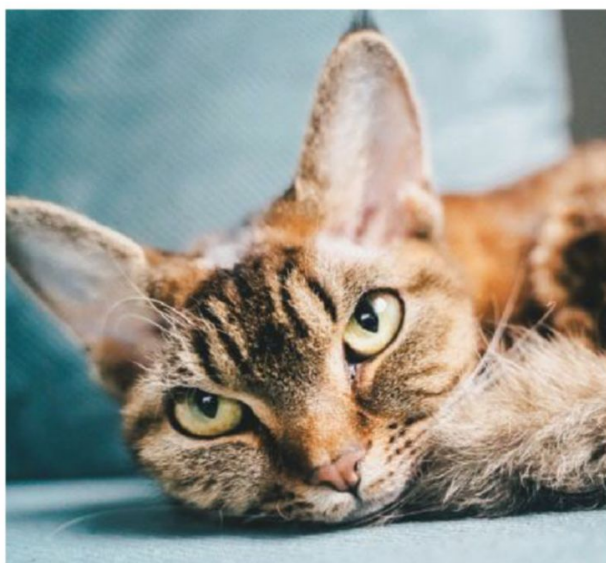
by sweeping away overlying sediments to reveal otherwise hidden igneous rock, the winds have managed to blur the boundary between yin and yang since the last shot was taken.

The new picture also captures shadowy fractures and pits that hint at large volumes of water ice still buried underneath the planet's surface, as well as numerous impact craters sur-

rounded by the detritus of their own explosive formation.

The Mars Express orbiter launched in 2003 and still provides fresh views of Earth's neighbor more than 20 years later, with each new image representing another clue in the enduring mystery of the Red Planet's long-lost, more Earth-like past.

—Joseph Howlett



Yana Iskayeva/Getty Images

was not involved in the study, says that “the authors made considerable efforts to rule out alternative explanations [for dogs’ motivation] such as attention, eye contact, object interest, and getting used to the situation.” It seemed they were trying to help.

But one question remained: Were cats failing to assist because they didn’t understand the situation—or because they lacked motivation?

To test this, the researchers added a final trial, replacing the sponge with food or a favorite toy. Cats then approached and indicated the object as often as dogs and children did.

“This brilliant study puts hard data to showing that cats aren’t mean but operate on a different evolutionary system,” says University of Pisa ethologist Elisabetta Palagi, who was not part of the study. Dogs and toddlers, she notes, are evolutionarily hardwired to treat another’s problem as their own. Cats, however, remain autonomous, understanding the situation without feeling compelled to intervene unless there’s a direct benefit for themselves. “They truly are the efficient specialists of the animal kingdom.” —Anirban Mukhopadhyay